



File permissions in Linux

Project description

In this project, I used Linux commands to view and manage file and directory permissions. I checked file details, interpreted permission strings, changed permissions for files and directories, and modified the permissions of a hidden file.

Check file and directory details

I used the `ls -l` command to display detailed information about files and directories.

`ls -l`

The command displays the file type, permissions, number of links, owner, group, file size, date and time of the last modification, and file name.

I also used the following command to display hidden files:

`ls -la`

The `-a` option displays all files, including hidden files.

```
researcher2@71dd6ffa453c:~$ pwd
/home/researcher2
researcher2@71dd6ffa453c:~$ ls
projects
researcher2@71dd6ffa453c:~$ cd projects
researcher2@71dd6ffa453c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 12:13 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 13:49 ..
-rw--w---- 1 researcher2 research_team  46 Sep 16 12:13 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 16 12:13 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Sep 16 12:13 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep 16 12:13 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 12:13 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 12:13 project_t.txt
researcher2@71dd6ffa453c:~/projects$
```



Describe the permissions string

A Linux permission string contains information about the file type and the permissions assigned to the owner, group, and other users.

For example:

```
-rwxr--r--
```

The first character indicates the file type. A `-` represents a regular file, while `d` represents a directory.

The next three characters represent the owner's permissions:

```
rwx
```

The following three characters represent the group's permissions:

```
r--
```

The final three characters represent the permissions for other users:

```
r--
```

The permissions are:

- `r` = read
- `w` = write
- `x` = execute
- `-` = permission not granted

Change file permissions

I granted read, write, and execute permissions to the user and removed write permissions from the group and other users for `project_k.txt`.



```
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 12:13 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 13:49 ..
-r--r----- 1 researcher2 research_team  46 Sep 16 12:13 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Sep 16 12:13 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Sep 16 12:13 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep 16 12:13 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 12:13 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 12:13 project_t.txt
researcher2@71dd6ffa453c:~/projects$ chmod u+x,g-w,o-w project_k.txt
researcher2@71dd6ffa453c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 12:13 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 13:49 ..
-r--r----- 1 researcher2 research_team  46 Sep 16 12:13 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Sep 16 12:13 drafts
-rwxr--r-- 1 researcher2 research_team  46 Sep 16 12:13 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep 16 12:13 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 12:13 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 12:13 project_t.txt
```

Change file permissions on a hidden file

```
researcher2@71dd6ffa453c:~/projects$ chmod g+r .project_x.txt
researcher2@71dd6ffa453c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 12:13 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 13:49 ..
-r--r----- 1 researcher2 research_team  46 Sep 16 12:13 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 16 12:13 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Sep 16 12:13 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep 16 12:13 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 12:13 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 12:13 project_t.txt
researcher2@71dd6ffa453c:~/projects$
```



Change directory permissions

```
researcher2@71dd6ffa453c:~/projects$ chmod g-x drafts
researcher2@71dd6ffa453c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 12:13 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 13:49 ..
-r--r----- 1 researcher2 research_team  46 Sep 16 12:13 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Sep 16 12:13 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Sep 16 12:13 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep 16 12:13 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 12:13 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 12:13 project_t.txt
```

Summary

In this project, I learned how to check and manage file and directory permissions in Linux. I used `ls -la` to view files, including hidden files, and learned how to interpret the 10-character permission string for the owner, group, and others. I also used `chmod` to add and remove permissions for files and directories. This project helped me understand how Linux permissions control access to files and directories and why they are important for security.